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Large Language Models Cannot Self-Correct Reasoning Yet

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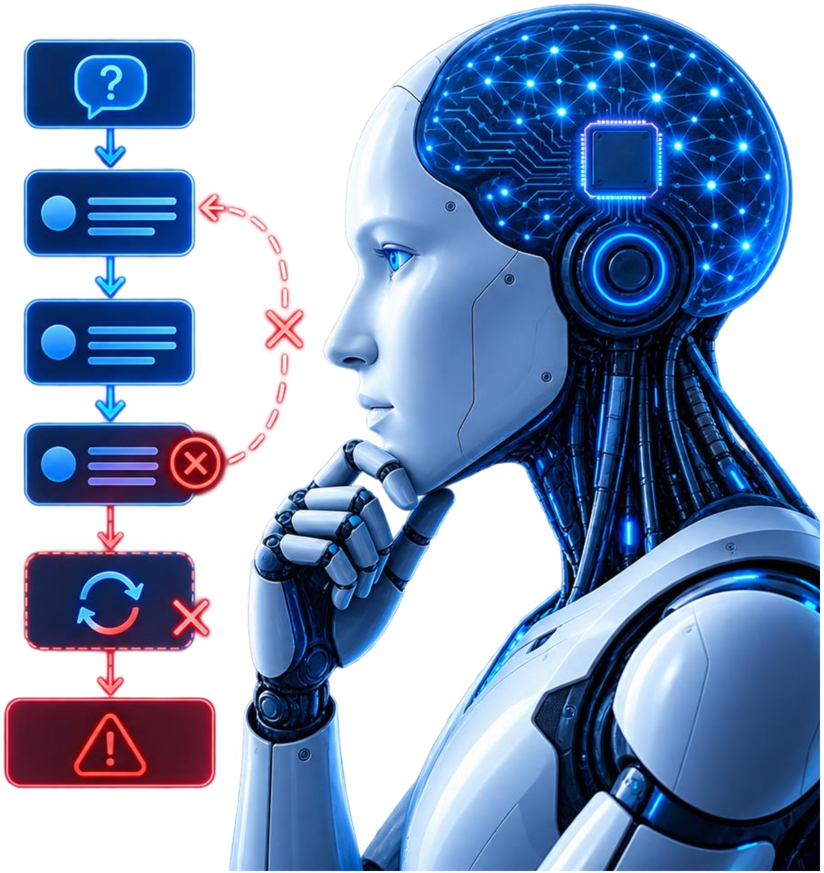

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OVERVIEW

Presentation Outline

Six sections tracing motivation to constructive future work

1 Introduction

Why multi-step reasoning fails and what self-correction promises.

2 Background & Related Work

RCI, Reflexion, Self-Refine, Debate and their evaluation flaws.

3 Methodology

GSM8K, CSQA, HotpotQA across GPT-3.5, GPT-4, Turbo, Llama-2.

4 Results

Accuracy declines after self-correction, net negative flips.

5 Analysis

Why gains vanish at equal compute and without feedback.

6 Conclusion & Future Work

External tools, verifiers, and fair evaluation standards.

Large Language Model (LLM)

1 **Next-Token Prediction**

Predicts the next token, one at a time.

2 **Pattern Learning**

Learns patterns and relationships from large amounts of text.

3 **Text Generation**

Generates human-like text based on the given input.

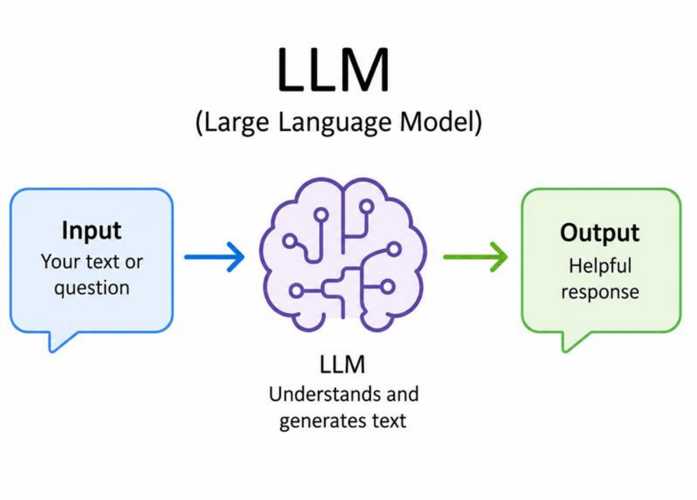

Figure-1:Large Language Model

LLM Self-Correction

Large Language Models (LLMs) generate text and solve problems by predicting likely word sequences, but their reasoning steps can contain errors.

1. **Initial response**
   Model answers the question on its own.
2. **Self-review**
   Model is asked to check its own answer and flag problems.
3. **Revised response**
   Model answers again, using its own feedback.

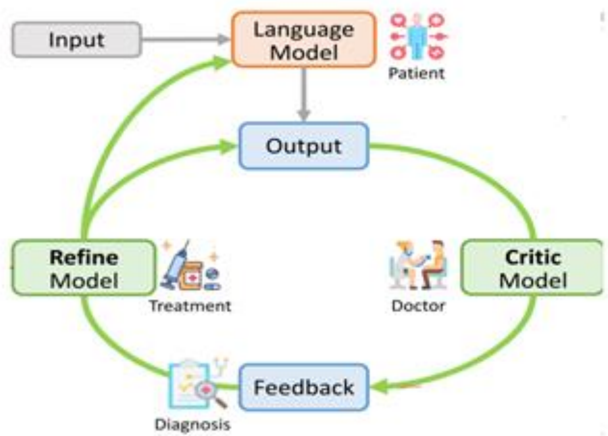

Figure-2: Large Language Model Self correction procedure

The Central Question

> "If an LLM possesses the ability to self-correct, why doesn't it simply offer the correct answer in its initial attempt?"
>
> Paraphrased from Huang et al., ICLR 2024, Section 1.

WHAT'S AHEAD

- Prior work and its evaluation gaps
- How the paper tests self-correction fairly
- The results: accuracy drops, not rises
- Why the drop happens, and what to do instead

Preview: across GSM8K, CommonSenseQA, and HotpotQA, the paper's answer is no. Left alone, LLMs do not reliably catch their own reasoning errors, and accuracy often falls after self-correction.

Key Terms for This Talk

Intrinsic self-correction

The model revises its own answer using only its own knowledge. No external feedback, no answer key, no tool.

Oracle feedback

Correction guided by a ground-truth label that tells the model whether its answer was already right.

Self-consistency

Sample several answers independently and take a majority vote. No critique step, just counting.

Multi-agent debate

Several copies of a model critique each other's answers over multiple rounds before settling on one.

Timeline of Claims

How 2023 was packed with self-correction hype

RCI
Kim et al.

Recursive critic

+7% claimed

Q1 2023

Reflexion
Shinn et al.

Verbal reinforcement

+11% claimed

Q2 2023

Self-Refine
Madaan et al.

Feedback refine

+10% claimed

Q3 2023

Debate
Du et al.

Multi-agent debate

+4% claimed

Q4 2023

WHAT THE TIMELINE HIDES

Oracle leakage
Only wrong answers are critiqued

Prompt trick
Rules withheld then re-added

Compute gap
Debate uses 3 to 6x calls vs single shot

Closed Loop vs Grounded Loop

Same model, different information: why internal reflection fails but external verification works

INTRINSIC : Closed Loop (Paper Focus)

1. Generate

LLM
Frozen Weights

2. Self-Critique

3. Revise Blind

ACCURACY CHANGE

-1.3% to -27.5%

No new facts. Shared models can hallucinate errors.

EXTERNAL : Grounded Loop (Tool-Assisted)

LLM
Drafts Code / Solution

Executes

Traceback

TOOL
Python / Oracle

ACCURACY CHANGE

+7.0% to +15.0%

Deterministic truth. Execution errors give precise correction signals.

Core Principle: A frozen model cannot act as an independent verifier of its own output without external grounding.

Forensic: Reported vs Controlled

Paired bars make inflation obvious

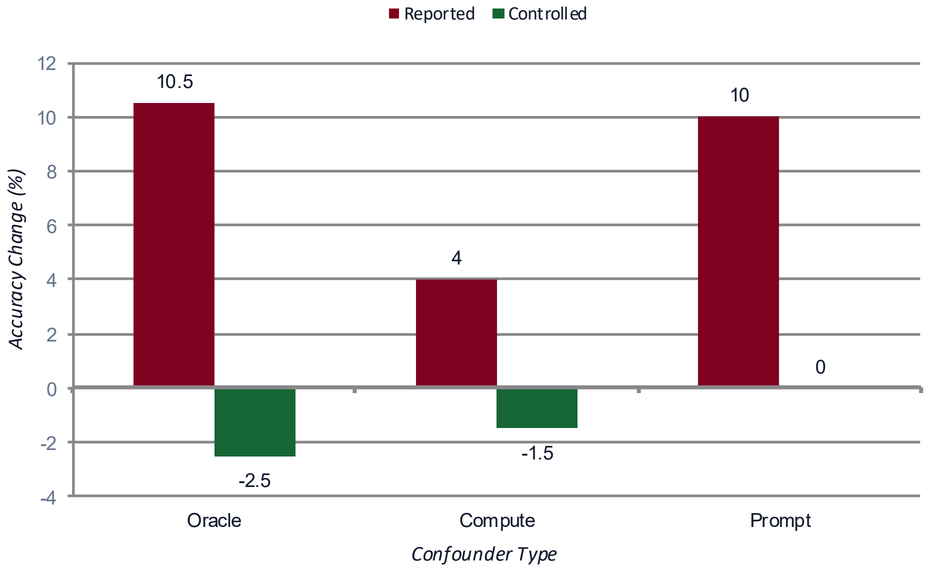

Figure-3: Accuracy Change (%) by Confounder Type

1. Oracle Leakage

+10.5% → -2.5%

Shield correct answers, only fix wrong ones.

2. Compute Gap

+4.0% → -1.5%

Debate loses to Self-Consistency at equal cost.

3. Prompt Trick

+10.0% → 0.0%

Weak start vs rule rich feedback.

Insights: Red collapses to green at or below zero. No confounder survives a fair test.

Benchmarks Used to Test Reasoning

Test a model's step-by-step logical and mathematical reasoning ability:

Dataset	Reasoning Task	Test Size	Prior Gain Claimed
GSM8K	Grade-school math word problems	1,319 questions	~7% (Kim et al., 2023)
CommonSenseQA	5-choice commonsense reasoning	1,221 questions (dev set)	~15% (Kim et al., 2023)
HotpotQA	Open-domain, multi-hop Q&A	100 questions	Sizeable gain (Shinn et al., 2023)

Why these three?

Each dataset was chosen because earlier self-correction papers reported their biggest improvements on it, giving this study the fairest and most challenging test of whether those gains hold up without oracle answer labels.

Models & Test Conditions

Minor changes in these settings can shift benchmark scores significantly:

Model	Settings Tested	Temperature	Sample Size
GPT-3.5-Turbo	Oracle + Intrinsic	1.0	Full test set
GPT-4	Oracle + Intrinsic	1.0	200 / 100 Qs
GPT-4-Turbo	Intrinsic only	0	200 Qs
Llama-2-70B-Chat	Intrinsic only	0	200 Qs

Max 2 rounds of self-correction were allowed per question, keeping inference cost comparable across models.

Prompting Procedures

Flow: intrinsic vs oracle regimes

1 Iniitial
Generate draft

2 Critique
Review logic

3 Revise
New reasoning

4 Iterate
Round 2 repeat

✓ With Oracle (Flawed)

- Verifier checks answer
- Tells model when wrong
- Correct answers untouched -> inflated

✗ Intrinsic (Realistic)

- No external signal
- Model judges itself blindly
- Flips correct -> wrong -> drops

Keeping the Evaluation Fair

Beyond accuracy numbers, the authors built two extra checks into their methodology:

Equal Inference Cost

Self-correction uses multiple model calls, so it is compared against self-consistency and multi-agent debate using the same number of responses, rather than against a single-call baseline.

Equal Prompt Effort

The initial-answer prompt and the feedback prompt are given the same level of detail, so gains can't come from simply hiding instructions in the correction step.

Multi-agent debate vs. self-consistency, and **prompt-design comparisons** both use this same fairness lens.

Intrinsic Self-Correction Fails

Main Finding: Performance declines after self-correction

CENTRAL EMPIRICAL FINDING

Intrinsic self-correction does NOT improve accuracy. Scores drop across all models and tasks.

1 Universal

All models

GPT-3.5, GPT-4, Turbo, Llama-2 all decline after R1 and R2.

↓

2 Task-Agnostic

All tasks

Math (GSM8K), logic (CSQA), QA (HotpotQA) all fail.

↓

3 Oracle Illusion

Prior artifact

Gains only with external labels guiding when to fix.

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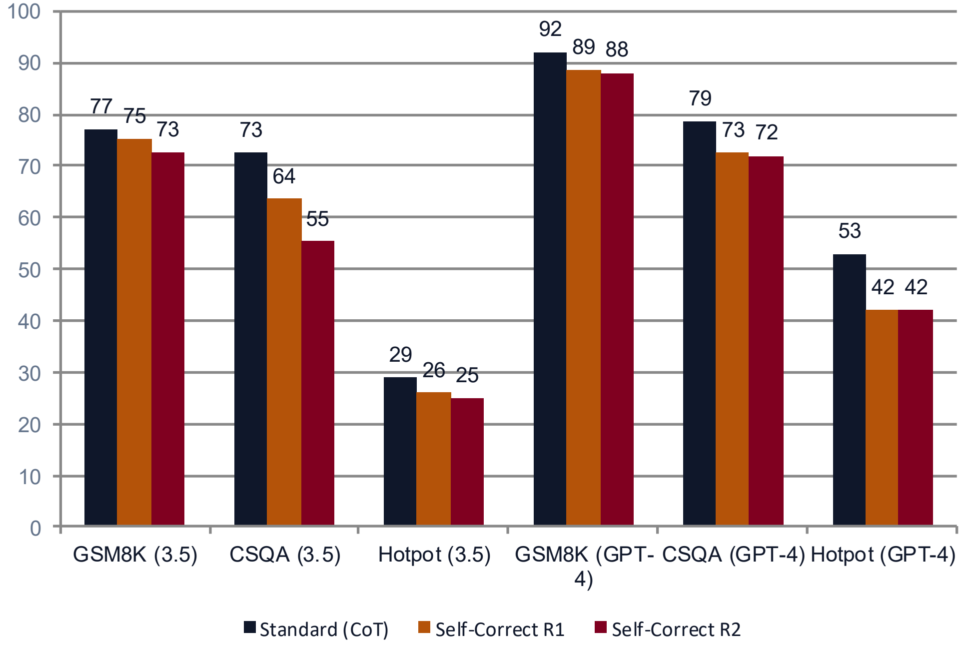
GPT-3.5 & GPT-4 Benchmark Results

Accuracy drops across rounds (Table 2)



Key Drops

GPT-3.5 on CSQA **-17.2%**

72.5% -> 55.3% (R2)

GPT-4 on HotpotQA **-11.0%**

53.0% -> 42.0%

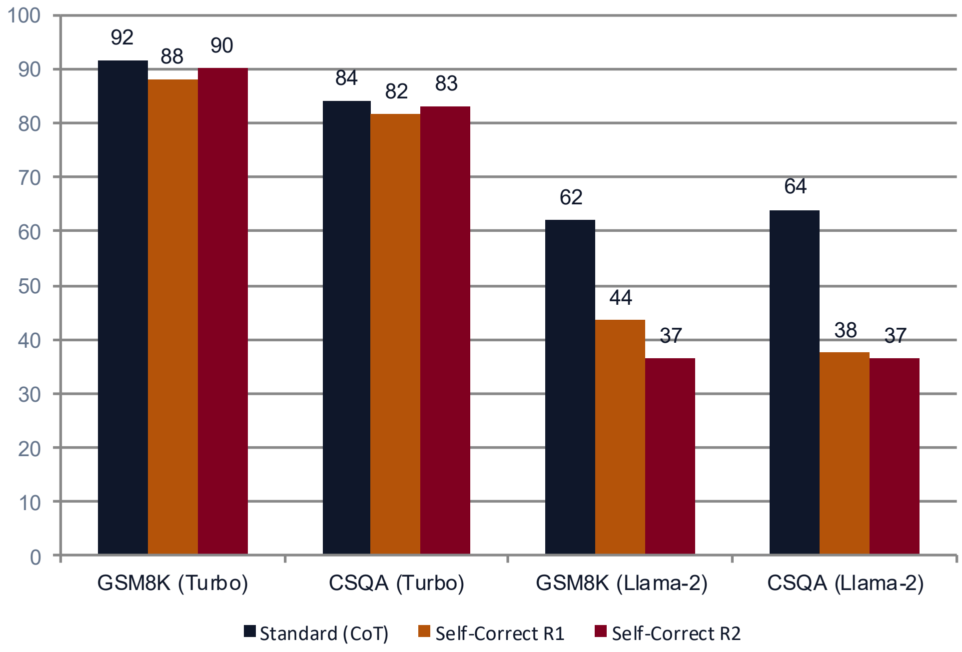
GPT-4 on GSM8K **-4.0%**

92.0% -> 88.0%

Table-1: GPT-3.5 and GPT-4 accuracy across two self-correction rounds

GPT-4-Turbo & Llama-2 Results

Failure generalizes across open and proprietary models



Llama-2 Collapse

GSM8K: 62.0% -> 36.5% (-25.5%)

CSQA: 64.0% -> 36.5% (-27.5%)

Open-weight models are highly prompt-compliant: they abandon correct answers when asked to "find flaws".

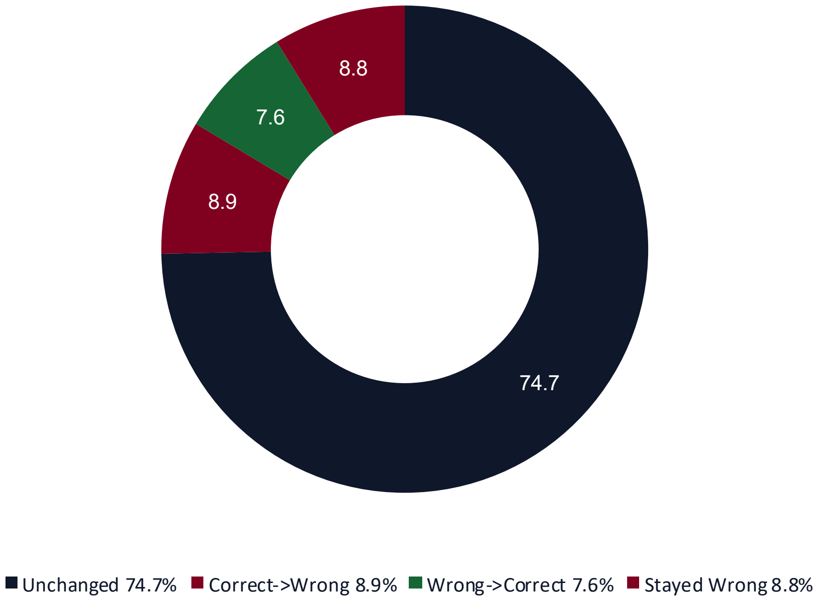
!

Extreme compliance vulnerability

Table-2: GPT-4-Turbo and Llama-2 results across self-correction rounds

Answer Transition Dynamics

Why accuracy drops: net-negative flips (Figure 1)



74.7% — **Unchanged**
Kept initial answer

8.9% — **Correct -> Wrong**
Harmful flip

7.6% — **Wrong -> Correct**
Beneficial fix

-1.3% — **Net Loss**
Harm > help

Figure-4: Answer transition dynamics after two self-correction rounds on GSM8K (GPT-3.5)

Why Does Self-Correction Fail?

What self-correction means, and the question this paper puts to the test.

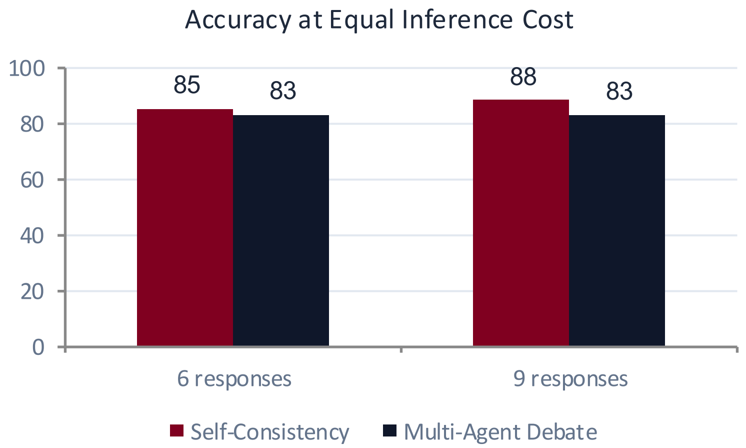
- Self-correction: an LLM revises its own answer using feedback on a prior output.
- This paper studies intrinsic self-correction, where the model corrects itself using only its own knowledge, with no external feedback or oracle labels.
- Central question: if a model can self-correct, why not answer correctly the first time?
- **Finding: without external feedback, LLMs largely fail to self-correct reasoning and often get worse.**

1. Generate answer
2. Self-review (no oracle)
3. Revised answer

Multi-Agent Debate ≈ Self-Consistency

At an equal number of model responses, debate offers no real advantage over simple majority voting (GSM8K).

Method	# Resp.	GSM8K Acc.
Standard Prompting	1	**76.7**
Self-Consistency	3	**82.5**
Multi-Agent Debate (r1)	6	**83.2**
Self-Consistency	6	**85.3**
Multi-Agent Debate (r2)	9	**83.0**
Self-Consistency	9	**88.2**



Takeaway: Debate is really "self-consistency with model-driven voting”. With the same number of responses, plain majority-vote self-consistency matches or beats it.

Case Study: The Gaslighting Effect

How critique prompts force correct reasoning into hallucination

GSM8K TASK **Terry eats 2 yogurts a day. They are sold in 4-packs at $5.00 per pack. How much does he spend on yogurts in 30 days?**

ROUND 1: CORRECT

$75

1. Total yogurts needed:
   30 days x 2 = 60 yogurts

2. 4-packs required:
   60 / 4 = 15 packs

3. Total expenditure:
   15 packs x $5 = $75

SELF-CRITIQUE PROMPT

"Review your answer and find flaws."

The Compliance Trap:

- Instruction-following bias
- Model assumes prompt implies an error occurred
- Model invents non-existent flaws to comply

ROUND 2: WRONG FLIP

$150

1. Model doubts previous logic

2. Hallucinates new rule:
   "Terry buys 1 pack daily"

3. Calculates 30 x $5 = $150
   (Flipped from 100% correct to 0%)

Insight: Asking an LLM to 'find flaws' acts as destructive noise because the model lacks an internal truth verifier.

Key Takeaways

1

Intrinsic Self-Correction Fails

Without external feedback or grounding, accuracy consistently drops after self-correction across GPT-3.5, GPT-4, GPT-4-Turbo, and Llama-2, on every reasoning benchmark tested.

2

Prior Gains Were Evaluation Artifacts

Reported improvements traced back to three confounders: oracle-label leakage (RCI, Reflexion), unfair compute parity (Debate vs. Self-Consistency), and sub-optimal initial prompts (Self-Refine).

3

Grounded Feedback Is Essential

A frozen model cannot reliably verify its own reasoning. Genuine gains require external signals, such as code execution, tool use, or trained verifiers, not more self-reflection.

CORE PRINCIPLE: A model that cannot generate the correct answer cannot reliably judge its own answer. Self-correction needs a verifier outside the generator.

Study Limitations & Methodological Boundaries

Scope and boundaries of this research

SCOPE LIMITATIONS

- **Reasoning-Specific :** Findings cover math, commonsense, and multi-hop QA, not style, tone, or safety self-correction.
- **Frozen Models Only :** Models were prompted, not fine-tuned, for self-correction; training-time approaches are untested.
- **2023 Model Snapshots :** GPT-3.5, GPT-4, GPT-4-Turbo, Llama-2-70B. Newer reasoning-focused models were not evaluated.

METHODOLOGICAL BOUNDARIES

- **Sample-Size Constraints :** HotpotQA evaluated on 100 questions to control API cost, limiting statistical resolution.
- **Non-Exhaustive Prompts :** Multiple feedback prompts were tested per model, but the space of possible phrasings is unbounded.
- **Closed-Book Setting :** No retrieval (RAG) or external documents were permitted during self-correction.

KEY INSIGHT: *These findings apply to intrinsic reasoning repair — not to style, safety, or preference alignment, where models may self-judge more reliably.*

Future Directions & Fair Evaluation Standards

Constructive paths forward

1 Tool-Augmented Execution

Route reasoning through Python REPLs, SQL engines, or solvers so an external, deterministic system, not the model itself, verifies correctness.

2 Learned Verifiers & PRMs

Train process-supervised reward models to judge intermediate reasoning steps, giving the corrector a grounded signal for where an error actually occurred.

3 Training-Time RL & Search

Invest in reinforcement learning and search (e.g., MCTS, test-time compute) so correction is learned during training, not bolted on as a prompting loop.

4 Fair Evaluation Standards

Report inference-cost-matched baselines against self-consistency, and disclose any oracle or ground-truth signal used to gate the correction loop.

Thank You

Open Discussions

We welcome questions from the professor and audience.

1. Why does asking a model to find flaws often make it hallucinate errors where none exist?
2. When should we rely on external tools like Python or verifiers instead of prompting alone?
3. How can we design a fair self-correction benchmark that avoids oracle leakage?